Guidance and Control of Unmanned Autonomous Vehicles

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UAV: Lateral Transfer Functions

- Variables of interest: ϕ, p, ψ, r
- Control inputs: δ_a, δ_r
- **Roll angle dynamics**:

$$\dot{\phi} = p + q \sin\phi \tan\theta + r \cos\phi \tan\theta$$

- In most of flight, $\theta \approx 0$, thus

$$\dot{\phi} = p + d_{\phi_1}; \quad d_{\phi_1} \triangleq q \sin\phi \tan\theta + r \cos\phi \tan\theta \implies \boxed{p = \dot{\phi} - d_{\phi_1}}$$

- On differentiating $\dot{\phi}$, we get

$$\begin{aligned}\ddot{\phi} &= \dot{p} + \dot{d}_{\phi_1} \\ &= \Gamma_1 pq - \Gamma_2 qr + \frac{1}{2}\rho V_a^2 S b \left[C_{p_0} + C_{p_\beta}\beta + C_{p_p}\frac{bp}{2V_a} + C_{p_r}\frac{br}{2V_a} + C_{p_{\delta_a}}\delta_a + C_{p_{\delta_r}}\delta_r \right] \\ &+ \dot{d}_{\phi_1}\end{aligned}$$

UAV: Lateral Transfer Functions

- On substituting for p, we get

$$\ddot{\phi} = \Gamma_1 pq - \Gamma_2 qr + \frac{1}{2}\rho V_a^2 Sb\left[C_{p_0} + C_{p_\beta}\beta + C_{p_p}\frac{b(\dot{\phi} - d_{\phi_1})}{2V_a} + C_{p_r}\frac{br}{2V_a} + C_{p_{\delta_a}}\delta_a\right.$$

$$\left. + C_{p_{\delta_r}}\delta_r\right] + \dot{d}_{\phi_1}$$

$$= \underbrace{\frac{1}{2}\rho V_a^2 SbC_{p_p}\frac{b}{2V_a}}_{-a_{\phi_1}}\dot{\phi} + \underbrace{\frac{1}{2}\rho V_a^2 SbC_{p_{\delta_a}}}_{a_{\phi_2}}\delta_a$$

$$+ \left\{\Gamma_1 pq - \Gamma_2 qr + \frac{1}{2}\rho V_a^2 Sb\left[C_{p_0} + C_{p_\beta}\beta - C_{p_p}\frac{bd_{\phi_1}}{2V_a} + C_{p_r}\frac{br}{2V_a} + C_{p_{\delta_r}}\delta_r\right] + \dot{d}_{\phi_1}\right\}$$

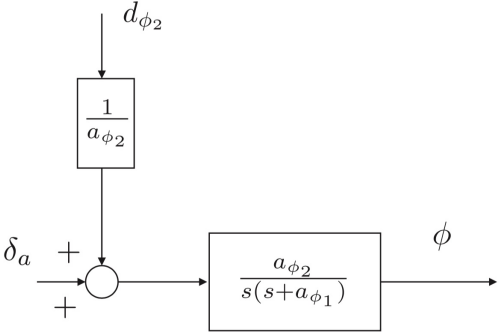
$$= -a_{\phi_1}\dot{\phi} + a_{\phi_2}\delta_a + d_{\phi_2}$$

- Can we write a transfer function for this differential equation now?

UAV: Lateral Transfer Functions

- In the Laplace domain,

$$\phi(s) = \frac{a_{\phi_2}}{s(s+a_{\phi_1})}\left(\delta_a(s) + \frac{1}{a_{\phi_2}} d_{\phi_2}(s)\right)$$



- **Course and Heading angle dynamics**: In coordinated turn without wind

$$\dot{\chi} = \frac{g}{V_g}\tan\phi$$

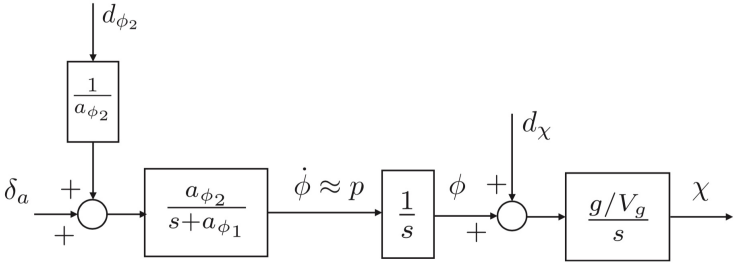
$$\dot{\chi} = \frac{g}{V_g}\phi + \frac{g}{V_g}\underbrace{(\tan\phi - \phi)}_{d_\chi}$$

$$\chi(s) = \frac{g}{V_g s}\left[\phi(s) + d_\chi(s)\right]$$

- With zero wind and $V_g \to V_a$,

$$\psi(s) = \frac{g}{V_g s}\left(\phi(s) + d_\chi(s)\right)$$

UAV: Lateral Transfer Functions



- **Sideslip angle**: In absence of wind, $v = V_a \sin\beta$
- For constant airspeed, $\dot{v} = V_a \cos\beta\dot{\beta}$

$$V_a \cos\beta\dot{\beta} = pw - ru + g\cos\theta\sin\phi + \frac{\rho V_a^2 S}{2m}\left(C_{Y_0} + C_{Y_\beta}\beta + C_{Y_p}\frac{bp}{2V_a} + C_{Y_r}r + C_{Y_{\delta_a}}\delta_a + C_{Y_{\delta_r}}\delta_r\right)$$

UAV: Lateral Transfer Functions

- With $\beta \approx 0$, $\cos\beta \approx 1$,

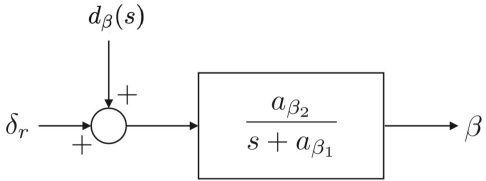
$$\dot{\beta} = -a_{\beta_1}\beta + a_{\beta_2}\delta_r + d_\beta$$

$$d_\beta = \frac{pw - ru + g\cos\theta\sin\phi}{V_a} + \frac{\rho V_a S}{2m}\left(C_{Y_0} + C_{Y_p}\frac{b}{2V_a}p + C_{Y_r}r + C_{Y_{\delta_a}}\delta_a\right)$$

$$a_{\beta_1} = \frac{\rho V_a S}{2m}C_{Y_\beta}, \quad a_{\beta_2} = \frac{\rho V_a S}{2m}C_{Y_{\delta_r}}$$

In the Laplace domain

$$\beta(s) = \frac{a_{\beta_2}}{s + a_{\beta_1}}\left[\delta_r(s) + d_\beta(s)\right]$$



Can we similarly obtain longitudinal transfer functions?

UAV: Longitudinal Transfer Functions

- Variables of interest: θ, q, V_a, $h = -p_d$
- Control inputs: Throttle and elevator surface deflections δ_t, δ_e
- **Pitch angle dynamics**:

$$\dot{\theta} = q\cos\phi - r\sin\phi = q + \underbrace{q(\cos\phi - 1) - r\sin\phi}_{d_{\theta_1}} \triangleq q + d_{\theta_1}$$

- Small d_{θ_1} for small ϕ
- On differentiating $\dot{\theta}$ w.r.t. time

$$\ddot{\theta} = \dot{q} + \dot{d}_{\theta_1}$$

- Using $\theta = \alpha + \gamma_a, \gamma_a = \gamma$

$$\ddot{\theta} = \Gamma_5 pr - \Gamma_6(p^2 - r^2) + \frac{\rho V_a^2 Sc}{2J_y}\left[C_{m_0} + C_{m_\alpha}\alpha + C_{m_q}\frac{b}{2V_a}q + C_{m_{\delta_e}}\delta_e\right] + \dot{d}_{\theta_1}$$

UAV: Longitudinal Transfer Functions

- Dynamics of pitch angle

$$
\begin{aligned}
\ddot{\theta} &= \Gamma_5 pr - \Gamma_6(p^2 - r^2) + \frac{\rho V_a^2 Sc}{2J_y}\left[C_{m_0} + C_{m_\alpha}(\theta - \gamma) + C_{m_q}\frac{b}{2V_a}(\dot{\theta} - d_{\theta_1}) + C_{m_{\delta_e}}\delta_e\right] \\
&\quad + \dot{d}_{\theta_1} \\
&= \underbrace{\frac{\rho V_a^2 Sc}{2J_y} C_{m_q}\frac{b}{2V_a}}_{-a_{\theta_1}}\dot{\theta} + \underbrace{\frac{\rho V_a^2 Sc}{2J_y} C_{m_\alpha}}_{-a_{\theta_2}}\theta + \underbrace{\frac{\rho V_a^2 Sc}{2J_y} C_{m_{\delta_e}}}_{a_{\theta_3}}\delta_e \\
&\quad + \Gamma_5 pr + \Gamma_6(r^2 - p^2) + \frac{\rho V_a^2 Sc}{2J_y}\left[C_{m_0} + C_{m_\alpha}(-\gamma) + C_{m_q}\frac{b}{2V_a}(-d_{\theta_1})\right] + \dot{d}_{\theta_1} \\
&= -a_{\theta_1}\dot{\theta} - a_{\theta_2}\theta + a_{\theta_3}\delta_e + d_{\theta_2}
\end{aligned}
$$

- How to write this differential equation in the Laplace domain?

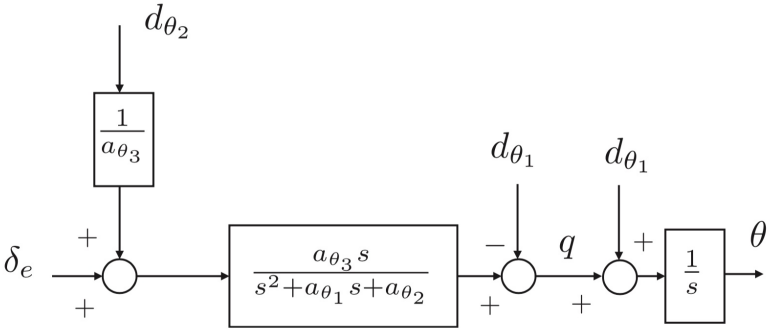
UAV: Longitudinal Transfer Functions

- In the Laplace domain,

$$\theta(s) = \frac{a_{\theta_3}}{s^2 + a_{\theta_1} s + a_{\theta_2}} \left[\delta_e(s) + \frac{1}{a_{\theta_3}} d_{\theta_2}(s) \right]$$

- For $C_{m_0} = 0$, $d_{\theta_2} = 0$.
- Also, $\dot{\theta} = q + d_{\theta_1}$

- For straight and level flight, $r = p = \phi = \gamma = 0$



UAV: Longitudinal Transfer Functions

- **Altitude**: For constant V_a, pitch angle directly influences climb rate of aircraft.
- Dynamics of altitude

$$\begin{aligned}\dot{h} &= u\sin\theta - v\sin\phi\cos\theta - w\cos\phi\cos\theta \\ &= V_a\theta + \underbrace{(u\sin\theta - V_a\theta) - v\sin\phi\cos\theta - w\cos\phi\cos\theta}_{d_h} = V_a\theta + d_h\end{aligned}$$

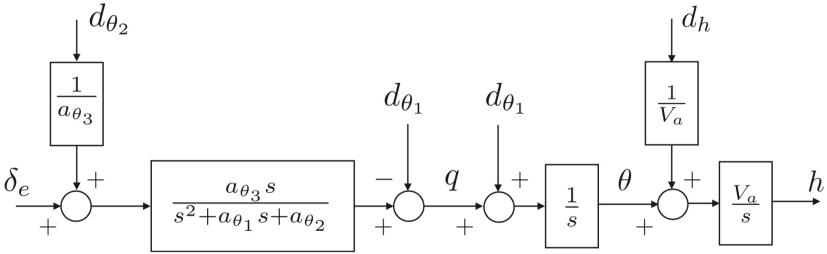
- In straight and level flight, $v \approx 0, w \approx 0, u \approx V_a, \phi \approx 0$ and θ small, and thus

$$d_h \approx 0$$

- For constant V_a with θ as input, in Laplace domain, one can write

$$sh(s) = V_a\theta + d_h(s) \implies h(s) = \frac{V_a}{s}\left(\theta + \frac{1}{V_a}d_h\right)$$

UAV: Longitudinal Transfer Functions



- For constant θ with V_a as input, in the Laplace domain, $h(s) = \frac{\theta}{s}\left(V_a + \frac{1}{\theta} d_h\right)$
- Either or both of these approaches can be used.
- How to obtain the relation among airspeed, throttle input, and pitch angle?
- **Inputs of interest**: Throttle input and pitch angle

UAV: Longitudinal Transfer Functions

- For zero wind speed, $V_a = \sqrt{u^2 + v^2 + w^2}$

$$\dot{V}_a = \frac{u\dot{u} + v\dot{v} + w\dot{w}}{V_a} = \dot{u}\cos\alpha\cos\beta + \dot{v}\sin\beta + \dot{w}\sin\alpha\cos\beta$$

$$\dot{V}_a = \dot{u}\cos\alpha + \dot{w}\sin\alpha + d_{V_1}, \quad \boxed{d_{V_1} \triangleq -\dot{u}(1-\cos\beta)\cos\alpha + \dot{v}\sin\beta - \dot{w}(1-\cos\beta)\sin\alpha}$$

- When $\beta = 0$, we have $d_{V_1} = 0$.
- On substituting for $\dot{u}$ and $\dot{w}$, we obtain

$$\begin{aligned}\dot{V}_a = &\cos\alpha\left[rv - qw - g\sin\theta + \frac{\rho V_a^2 S}{2m}\left(C_X(\alpha) + C_{X_q}(\alpha)\frac{cq}{2V_a} + C_{X_{\delta_e}}(\alpha)\delta_e\right)\right.\\ &\left.+\frac{\rho S_{\text{prop}} C_{\text{prop}}}{2m}\left\{(k_{\text{motor}}\delta_t)^2 - \rho V_a^2\right\}\right] + \sin\alpha\left[qu - pv + g\cos\theta\cos\phi\right.\\ &\left.+\frac{\rho V_a^2 S}{2m}\left(C_Z(\alpha) + C_{Z_q}(\alpha)\frac{cq}{2V_a} + C_{Z_{\delta_e}}(\alpha)\delta_e\right)\right] + d_{V_1}\end{aligned}$$

UAV: Longitudinal Transfer Functions

- Recall aerodynamic coefficients for UAV

$$C_X(\alpha) \triangleq -C_D(\alpha)\cos\alpha + C_L(\alpha)\sin\alpha, \qquad C_Z(\alpha) \triangleq -C_D(\alpha)\sin\alpha - C_L(\alpha)\cos\alpha$$

$$C_{X_q}(\alpha) \triangleq -C_{D_q}\cos\alpha + C_{L_q}\sin\alpha, \qquad C_{Z_q}(\alpha) \triangleq -C_{D_q}\sin\alpha - C_{L_q}\cos\alpha$$

$$C_{X_{\delta_e}}(\alpha) \triangleq -C_{D_{\delta_e}}\cos\alpha + C_{L_{\delta_e}}\sin\alpha, \qquad C_{Z_{\delta_e}}(\alpha) \triangleq -C_{D_{\delta_e}}\sin\alpha - C_{L_{\delta_e}}\cos\alpha$$

$$C_L(\alpha) = C_{L_0} + C_{L_\alpha}\alpha, \qquad C_D(\alpha) = C_{D_0} + C_{D_\alpha}\alpha$$

- Using linear approximation and performing algebraic simplifications

$$\begin{aligned}\dot{V}_a =& rV_a\cos\alpha\sin\beta - pV_a\sin\alpha\sin\beta - g\cos\alpha\sin\theta + g\sin\alpha\cos\theta\cos\phi \\ &+ \frac{\rho V_a^2 S}{2m}\left(-C_{D_0} - C_{D_\alpha}\alpha - C_{D_q}\frac{c}{2V_a}q + C_{D_{\delta_e}}\delta_e\right) \\ &+ \frac{\rho S_{\text{prop}} C_{\text{prop}}}{2m}\left\{(k_{\text{motor}}\delta_t)^2 - \rho V_a^2\right\}\cos\alpha + d_{V_1}\end{aligned}$$

UAV: Longitudinal Transfer Functions

- On simplifying the previous expression

$$
\begin{aligned}
\dot{V}_a =& (rV_a \cos\alpha - pV_a \sin\alpha)\sin\beta - g\sin(\theta-\alpha) - g\sin\alpha\cos\theta(1-\cos\phi) \\
&+ \frac{\rho V_a^2 S}{2m}\left(-C_{D_0} - C_{D_\alpha}\alpha - C_{D_q}\frac{c}{2V_a}q + C_{D_{\delta_e}}\delta_e\right) \\
&+ \frac{\rho S_{\text{prop}} C_{\text{prop}}}{2m}\left\{(k_{\text{motor}}\delta_t)^2 - \rho V_a^2\right\}\cos\alpha + d_{V_1} \\
=& -g\sin\gamma + \frac{\rho V_a^2 S}{2m}\left(-C_{D_0} - C_{D_\alpha}\alpha - C_{D_q}\frac{c}{2V_a}q + C_{D_{\delta_e}}\delta_e\right) \\
&+ \frac{\rho S_{\text{prop}} C_{\text{prop}}}{2m}\left\{(k_{\text{motor}}\delta_t)^2 - V_a^2\right\} + d_{V_2}
\end{aligned}
$$

$$
\begin{aligned}
d_{V_2} =& (rV_a \cos\alpha - pV_a \sin\alpha)\sin\beta - g\sin\alpha\cos\theta(1-\cos\phi) \\
&+ \frac{\rho S_{\text{prop}} C_{\text{prop}}}{2m}\left\{(k_{\text{motor}}\delta_t)^2 - V_a^2\right\}(\cos\alpha - 1) + d_{V_1}
\end{aligned}
$$

- In a level flight, $d_{V_2} = 0$.

UAV: Longitudinal Transfer Functions

- **Dynamics of** V_a: Nonlinear in both δ_t and V_a
- How to deal with this issue to obtain the transfer function then?
- Define deviations from trim values as $\bar{V}_a = V_a - V_a^\star$, $\bar{\theta} = \theta - \theta^\star$, $\bar{\delta}_t = \delta_t - \delta_t^\star$
- Linearization around level wings, constant altitude $(\gamma^\star = 0)$

$$\dot{\bar{V}}_a = -\underbrace{g\cos(\theta^\star - \alpha^\star)}_{a_{V_3}}\bar{\theta} + \underbrace{\left\{\frac{\rho V_a^\star S}{m}\left[-C_{D_0} - C_{D_\alpha}\alpha^\star - C_{D_{\delta_e}}\delta_e^\star\right] - \frac{\rho S_{\text{prop}} C_{\text{prop}}}{m} V_a^\star\right\}}_{-a_{V_1}}\bar{V}_a$$

$$+ \underbrace{\frac{\rho S_{\text{prop}} C_{\text{prop}} k_{\text{motor}}^2 \delta_t^\star}{m}}_{a_{V_2}}\bar{\delta}_t + d_{V_2}$$

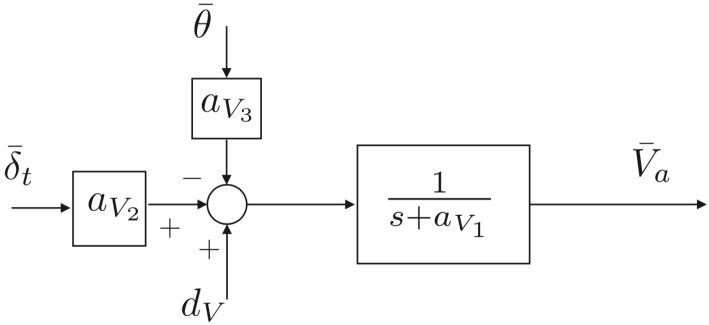
$$= -a_{V_1}\bar{V}_a + a_{V_2}\bar{\delta}_t - a_{V_3}\bar{\theta} + d_V$$

- d_V includes d_{V_2} and HOT in the Taylor's series

UAV: Longitudinal Transfer Functions

- In the Laplace domain,

$$\bar{V}_a(s) = \frac{1}{s + a_{V_1}} \left[a_{V_2} \bar{\delta}_t(s) - a_{V_3} \bar{\theta}(s) + d_V(s) \right]$$



Unmanned Autonomous Vehicles: Autopilot

Autopilot

A system used to guide an aircraft without the assistance of a pilot.

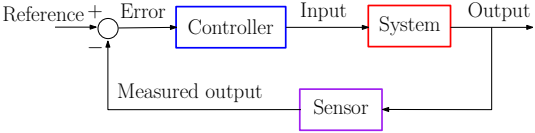
- Manned aircraft
  - A single-axis wing-leveling autopilot
  - A full flight control system to control position and attitude during various phases of flight
- Unmanned vehicle
  - Complete control of aircraft during all phases of flight
- **Autopilot**: Suitable for available sensor and computational resources

PC:

https://gadgetsdeal.in

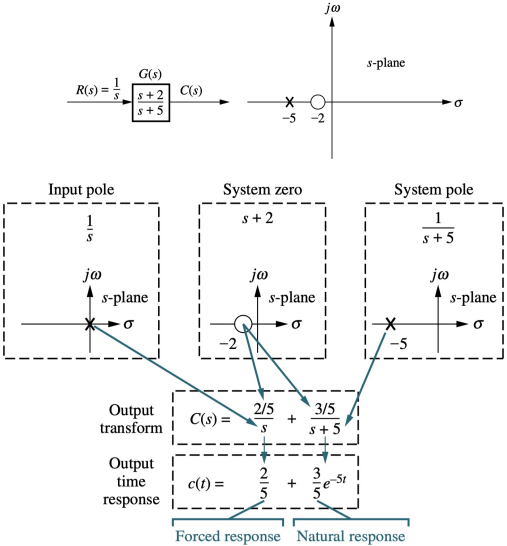
Unmanned Autonomous Vehicles: Feedback System

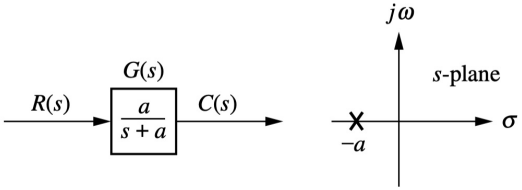
- Feedback control system



- **Transfer function**: A ratio of the Laplace transforms of output to that of input, with zero initial conditions.
- **Poles**:
  - Value of s that causes transfer function to become infinite
  - Any roots of the denominator of the transfer function, common to roots of the numerator.
- **Zeros**:
  - Value of s that causes transfer function to become zero
  - Any roots of the numerator of transfer function, common to the roots of

Unmanned Autonomous Vehicles: Poles and Zeros



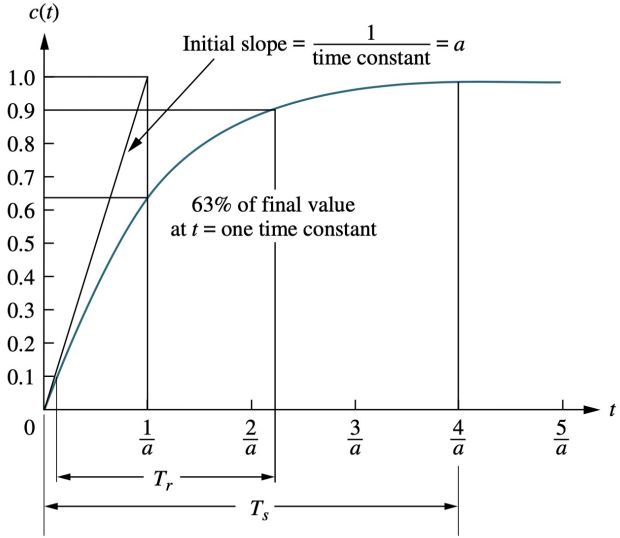


- System response for step input:

$$C(s) = G(s)R(s) = \frac{a}{s(s+a)}$$

- In time domain: $c(t) = 1 - e^{-at}$
- Time constant: $1/a$

Unmanned Autonomous Vehicles: Response of first order system



Rise time (T_r)

Time for the waveform to go from 0.1 to 0.9 of its final value

$$T_r = \frac{2.2}{a}$$

Settling Time (T_s)

Time for the response to reach, and stay within, 2% of its final value

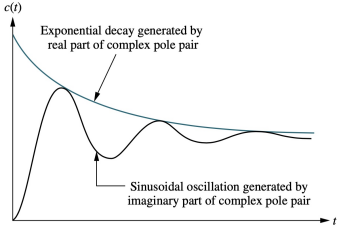
$$T_s = \frac{4}{a}$$

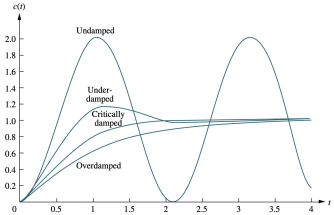
Unmanned Autonomous Vehicles: Response of second order system

- General second-order system with unity DC gain

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

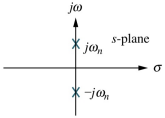
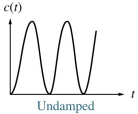
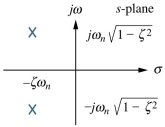
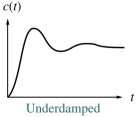
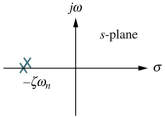
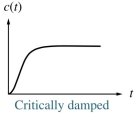
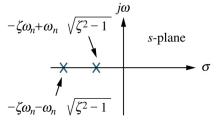
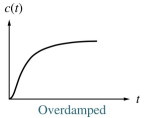
- Natural frequency: Frequency of oscillation of system without damping
- Damping ratio: Ratio of exponential decay frequency to natural frequency





What would be the fastest response without overshoot? Critically damped one

Unmanned Autonomous Vehicles: Response of second order system

ζ	Poles	Step response
0		
$0 < \zeta < 1$		
$\zeta = 1$		
$\zeta > 1$		

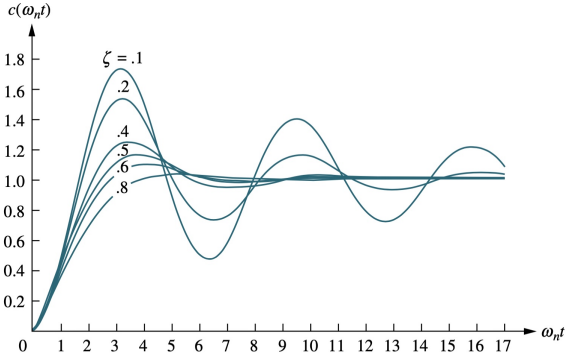
Unmanned Autonomous Vehicles: Response of second order system

- Underdamped second-order system

$$G(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

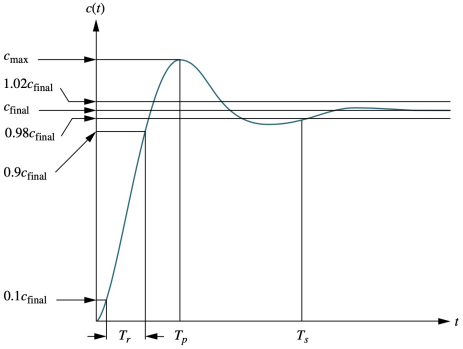
- Output for step input

$$C(s) = \frac{\omega_n^2}{s(s^2 + 2\zeta\omega_n s + \omega_n^2)}$$



$$c(t) = 1 - \frac{1}{\sqrt{1-\zeta^2}} e^{-\zeta\omega_n t} \cos\left(\underbrace{\omega_n\sqrt{1-\zeta^2}}_{\omega_d} t - \phi\right), \ \phi = \tan^{-1}\left(\frac{\zeta}{\sqrt{1-\zeta^2}}\right)$$

Unmanned Autonomous Vehicles: Response of second order system



- **Rise time (T_r)**: Time required for waveform to go from 0 to 1 (0.1 to 0.9) of the final value.
- **Peak time (T_P)**: Time required to reach first, or maximum, peak.
- **Settling time (T_s)**: Time required for the transient's damped oscillations to reach and stay within 2% of the steady-state value.
- **Percent overshoot ($\%OS$)**: Amount that waveform overshoots steady-state or final value at peak time, as % of the steady-state value.

These times represent the speed of transient response of overall system.

Unmanned Autonomous Vehicles: Response of second order system

- How to compute the peak time?
- By differentiating $c(t)$ and finding the first zero crossing after $t = 0$.

$$\boxed{T_p = \frac{\pi}{\omega_n\sqrt{1-\zeta^2}} = \frac{\pi}{\omega_d}}$$

- How to compute the peak overshoot?

$$\boxed{\%OS = \left(\frac{c_{\text{max}} - c_{\text{final}}}{c_{\text{final}}}\right) 100 = 100\, e^{-\left(\pi\zeta/\sqrt{1-\zeta^2}\right)}}$$

- How to compute the settling time?

$$\boxed{T_s = \frac{\ln 0.002\sqrt{1-\zeta^2}}{\zeta\omega_n} \approx \frac{4}{\zeta\omega_n} = \frac{4}{\sigma_d}, \quad \ln 0.002\sqrt{1-\zeta^2} \in (3.91, 4.74)}$$

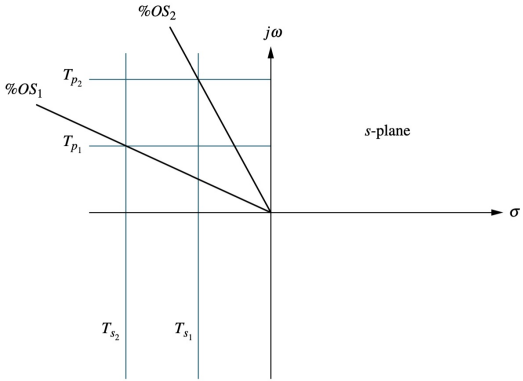
Unmanned Autonomous Vehicles: Response of second order system

- Rise time ($\zeta = 0.5$): $T_R = \dfrac{1.8}{\omega_n}$
- Approximations using curve-fitting techniques:

$$\omega_n T_r = 1.76\zeta^3 - 0.417\zeta^2 + 1.039\zeta + 1$$

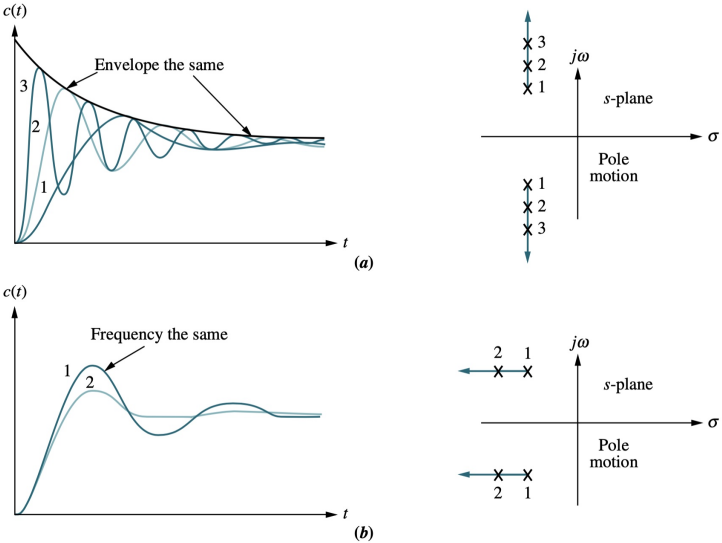
$$\zeta = 0.115\left(\omega_n T_r\right)^3 - 0.883\left(\omega_n T_r\right)^2 + 2.504\left(\omega_n T_r\right) - 1.738$$

- T_p: $1/\propto$ Imaginary part of pole
- T_s: $1/\propto$ Real part of pole

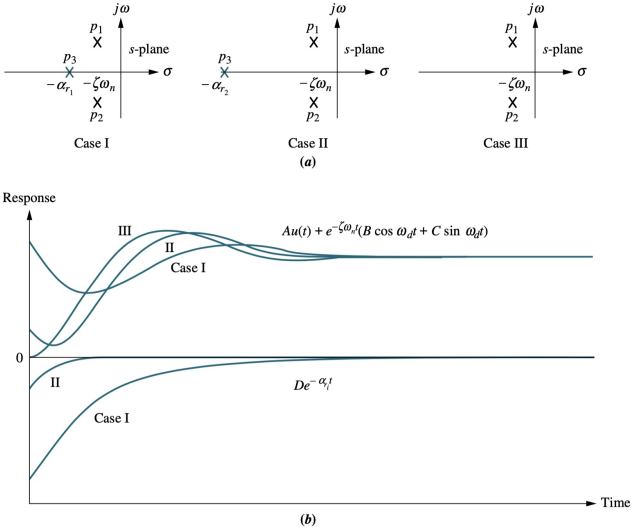


Percent overshoot: A function of ζ

Unmanned Autonomous Vehicles: Feedback System



Unmanned Autonomous Vehicles: Feedback System with 3 Poles



- How much farther from the dominant poles does the third pole have to be for its effect on the second-order response to be negligible?
- At least 5 times
- Additional pole increases T_r, that is, slows down the system.

Text/References

References

- Randal Beard and Timothy W. McLain, "Small Unmanned Aircraft: Theory and Practice", Princeton University Press, 2012
- Norman Nise, "Control System Engineering", Wiley, Eighth Edition, 2020.

Thank you for your attention !!!